

Experiment - 2

AIM - Convert the ER diagram into a relational schema. Write Complete MySQL, CREATE TABLE Statements with primary keys, foreign keys, NOT NULL, UNIQUE, ON DELETE CASCADE and SETNULL constraints. Insert sample data and demonstrate referential integrity violation.

RELATIONAL SCHEMA

Customer (

CustomerID PK,
Name,
Email UNIQUE,
Phone No,
DOB

)

Seller (

SellerID PK,
Seller Name,
Phone No UNIQUE

)

Product (

P. ProductID Fk,
CategoryID Fk,
Price,
Image

)

Address (

AddressID PK,
CustomerID Fk,
Pincode,
State,
City

)

category (

CategoryID PK,
Category Name UNIQUE,
Description

)

Electronic (

ProductID PK,
Fk

)

Fashion (

ProductID Pk

Fk

)

Orders (

OrderID Pk,

CustomerID Fk,

Order Date,

Order Status

)

Payment

PaymentID Pk,

orderID Fk UNIQUE,

Payment Date,

Amount,

Payment Method,

Status

)

Books (

ProductID Pk,

Fk,

SellerID Fk

)

Order-item (

OrderID Pk/Fk,

item-seq-no Pk,

ProductID Fk,

Quantity,

Unit-Price

)

Delivery (

deliveryID Pk,

orderID Fk UNIQUE,

Courier Name,

Status,

Delivery Date

)

SOURCE CODE -

```
CREATE DATABASE Ecommerce DB;  
USE Ecommerce DB;
```

```
CREATE TABLE Customer C
```

```
CustomerID INT Primary key Auto-Increment,  
Name Varchar(100) NOT NULL,  
Email Varchar(100) NOT NULL UNIQUE,  
PhoneNo Varchar(15) NOT NULL,  
DOB DATE NOT NULL
```

```
);
```

```
CREATE TABLE Address C
```

```
Address ID INT Primary key Auto-increment,  
CustomerID INT NOT NULL,  
Pincode Varchar(10) NOT NULL,  
State Varchar(50) NOT NULL,  
City Varchar(50) NOT NULL,  
FOREIGN KEY (CustomerID) references Customer  
(CustomerID) ON DELETE CASCADE
```

```
);
```

```
CREATE TABLE Category C
```

```
Category ID INT Primary key Auto-increment,  
Category Name Varchar(100) NOT NULL UNIQUE,  
Description Varchar(255)
```

```
);
```

CREATE TABLE SELLER(

SellerID INT Primary key auto-increment,

SellerName Varchar(100) NOT NULL,

Phone No Varchar(15) NOT NULL UNIQUE

);

CREATE TABLE Product (

ProductID INT Primary key auto-increment,

Category ID INT NOT NULL,

Price Decimal (10,2) NOT NULL,

Image Varchar(255),

Foreign key (CategoryID) references Category

(CategoryID) ON DELETE CASCADE ~

);

CREATE TABLE orders(

OrderID INT Primary key auto-increment,

CustomerID INT NOT NULL,

Order Date DATE NOT NULL,

Order Status Varchar(50) NOT NULL,

Foreign key (CustomerID) references Customer

(CustomerID) ON DELETE CASCADE)

);

CREATE TABLE OrderItem (

OrderID INT NOT NULL,

item-seq-no INT NOT NULL,

ProductID INT NOT NULL,

Quantity INT NOT NULL,

Unit Price DECIMAL (10,2) NOT NULL,

Primary key (OrderID, item-seq-no),

Foreign key (OrderID) references Orders (OrderID)
ON DELETE CASCADE

Foreign key (ProductID) references Product (ProductID)
ON DELETE CASCADE

);

CREATE TABLE Payment (

PaymentID INT Primary key auto-increment,

OrderID INT NOT NULL UNIQUE,

Payment Date Date NOT NULL,

Amount Decimal (10,2) NOT NULL,

Payment method Varchar (30) NOT NULL,

Status Varchar (30) NOT NULL,

Foreign key (OrderID) references order (OrderID)
ON DELETE CASCADE

);

CREATE TABLE Delivery (

DeliveryID INT Primary key auto-increment

orderID INT NOT NULL UNIQUE,

Courier Name Varchar(100) NOT NULL,

Status Varchar(30) NOT NULL,

Delivery Date DATE,

Foreign key (order ID) references order(orderID)

ON DELETE CASCADE.

);

Inserting Sample Data -

Insert into Customer (Name, Email, Phone No, DOB) VALUES
("Sahil", "Sahil@gmail.com", "9039105025", "2005-05-15"),
("Arjun", "Arjun@gmail.com", "9336677909", "2004-08-20");

insert into category (category Name, Description) VALUES

("Electronics", "Electronic Products"),

("Fashion", "Fashion Product"), ("Books", "Books")

insert into ~~seller~~ (seller Name, Phone No) VALUES

("Tech store", "90000000001"),

("Book Store", "90000000002");

insert into Product (categoryID, Price, image) VALUES
(1, 50000, "Laptop.jpg"), (2, 999, "shirt.png"),
(3, 599, "book.jpg");

Insert into Electronic Values (1);

Insert into Fashion Values (2);

Insert into Books Values (3, 2);

Insert into Address (customerID, Pincode, State, City) VALUES

(1, "208001", "Uttar Pradesh", "Kanpur"),

(2, "110001", "Delhi", "Delhi");

Insert into Orders (customerID, orderDate, OrderStatus) VALUES

(1, "2026-08-19", "Placed");

(2, "2026-08-19", "Shipped");

Insert into orderItem VALUES

(1, 1, 1, 1, 50000), (2, 1, 2, 2, 999);

Insert into Payment (Order ID, Payment Date, Amount,
Payment Method, Status) VALUES

(1, "2026-08-19", 50000, "UPI", "SUCCESS"),

(2, "2026-08-19", 1998, "Card", "Success");

Insert into Delivery (Order ID, Courier Name, Status, Delivery Date)
VALUES

(1, "Delivery", "Delivered", "2026-08-20"),

(2, "DTDC", "Pending", NULL);

Referential integrity violations -

Insert into order (CustomerID, orderDate, OrderStatus)
VALUES
(999, "2026-08-19", "Placed");

OUTPUT :-

ERROR: cannot add or update a child row:
a foreign key constraint fails.

Unique Constraint Violation

Insert into Customer (Name, Email, PhoneNO, DOB) VALUES
("TEST", "Sahil@gmail.com", "9999999999", "2025-01-01");

OUTPUT:-

ERROR: Duplicate entry "Sahil@gmail.com"
for key 'Customer.Email'

Select * from Customer;

Select * from Product;

Select * from Orders;

Select * from Orderitem;

ON DELETE SET NULL

DELETE from Seller where SellerID = 2;

Select * from Books;

ON DELETE CASCADE

DELETE from Customer where CustomerID = 2;

Select * from Order where CustomerID = 2;

OUTPUT -

Customer ID	Name	Email	Phone No	DOB
1	Shardendu	Shardendu000@gmail.com	8853936527	2008-10-08
2	Arjun	Arjun@gmail.com	9336677989	2004-08-20

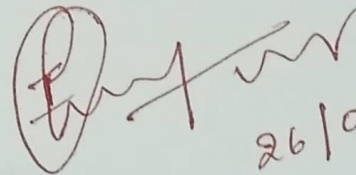
Product ID	Category ID	Price	Image
1	1	50000.00	Laptop. Png
2	2	999.00	Shirt. Png
3	3	599.00	book. Png

Order ID	Customer ID	Order Date	order Status
1	1	2026-08-19	Placed
2	2	2026-08-19	Shipped

Order ID	item-seq-no	Product ID	Quantity	Unit-Price
1	1	1	1	50000.00
2	1	2	2	999.00

Product ID	Seller ID
3	NULL

Empty Set


26/08/26